



UPPSALA UNIVERSITET

NUMERICAL LINEAR ALGEBRA · 1TD452

Signal and Image Processing

Mini-project 1: Direct Methods

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1 Introduction

Many problems in signal and image processing have the same basic form. We want to recover an unknown signal x , but we can only observe a distorted and noisy version of it,

$$b = Ax + e. \quad (1)$$

Here b is the observed data, A is the *forward operator* that describes how the measurement system (for example a blurring camera) distorts the signal, and e is unknown noise.

The natural first attempt is to solve the least-squares problem $\min_x \|Ax - b\|_2^2$. But in many applications the matrix A is ill-conditioned. This means that a small changes in b cause large changes in the solution.

What we often do is apply *regularization*. We add a penalty term that rewards solutions we consider reasonable. This gives the regularized problem

$$\min_x \|Ax - b\|_2^2 + \lambda^2 \|Lx\|_2^2. \quad (2)$$

The parameter $\lambda > 0$ controls the strength of the regularization. The matrix L encodes what we know about the signal in advance and depends on the application. For example, $L = I$ favours solutions of small size, while a difference operator favours smooth solutions.

2 Method

2.1 Tikhonov regularization and the SVD

The singular value decomposition (SVD) is a useful tool for understanding least-squares problems. We splits A into simple building blocks.

$$A = U\Sigma V^T, \quad (3)$$

U and V are orthogonal matrices and Σ is a diagonal matrix holding the singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ of A . We denote the columns of U and V by u_i and v_i .

Using the SVD, the ordinary least-squares solution of $\min_x \|Ax - b\|_2^2$ can be written as

$$x_{\text{LS}} = \sum_{i=1}^n \frac{u_i^T b}{\sigma_i} v_i. \quad (4)$$

This formula shows exactly where the trouble comes from. When A is ill-conditioned, some σ_i are tiny, and dividing by them blows up the noise in b . To prevent this we use tikhonov regularization and add a penalty on the size of x

$$\min_x f(x) := \|Ax - b\|_2^2 + \lambda^2 \|x\|_2^2, \quad (5)$$

which is Equation 2 with $L = I$. Expanding f gives

$$f(x) = (Ax - b)^T (Ax - b) + \lambda^2 x^T x = x^T A^T A x - 2x^T A^T b + b^T b + \lambda^2 x^T x. \quad (6)$$

Since f is strictly convex, x is the minimizer exactly when the gradient vanishes, $\nabla f(x) = 2(A^T A + \lambda^2 I)x - 2A^T b = 0$. This is the same as the *normal equations*

$$(A^T A + \lambda^2 I)x = A^T b. \quad (7)$$

The matrix $A^T A + \lambda^2 I$ is positive definite and hence invertible, so the normal equations have exactly one solution, $x_{\text{RLS}} = (A^T A + \lambda^2 I)^{-1} A^T b$.

To see what the regularization actually does, we insert the SVD into the normal equations and simplify using $U^T U = I$ and $V^T V = V V^T = I$:

$$\begin{aligned}
(A^T A + \lambda^2 I)x_{\text{RLS}} &= A^T b \\
V \Sigma^T U^T U \Sigma V^T + \lambda^2 V V^T x_{\text{RLS}} &= V \Sigma^T U^T b \\
V(\Sigma^T \Sigma + \lambda^2 I)V^T x_{\text{RLS}} &= V \Sigma^T U^T b \\
V^T V(\Sigma^T \Sigma + \lambda^2 I)V^T x_{\text{RLS}} &= V^T V \Sigma^T U^T b \\
\underbrace{(\Sigma^T \Sigma + \lambda^2 I)}_B \underbrace{V^T x_{\text{RLS}}}_y &= \Sigma^T U^T b \\
B = \begin{bmatrix} \sigma_1^2 + \lambda^2 & & & \\ & \sigma_2^2 + \lambda^2 & & \\ & & \ddots & \\ & & & \sigma_n^2 + \lambda^2 \end{bmatrix} & \\
\Rightarrow y = B^{-1} \Sigma^T U^T b & \\
= \begin{bmatrix} \frac{1}{\sigma_1^2 + \lambda^2} & & & \\ & \frac{1}{\sigma_2^2 + \lambda^2} & & \\ & & \ddots & \\ & & & \frac{1}{\sigma_n^2 + \lambda^2} \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix} \begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_n^T \end{bmatrix} b &
\end{aligned} \quad (8)$$

Because B^{-1} and Σ are both diagonal, each component of y can be read off directly:

$$y_i = \frac{\sigma_i}{\sigma_i^2 + \lambda^2} u_i^T b, \quad i = 1, \dots, n, \quad (9)$$

and since V is orthogonal we recover the solution as $x_{\text{RLS}} = Vy$.

We assume A has full rank, so every $\sigma_i > 0$. We can then split the coefficient and introduce the *filter factor* $\phi_i(\lambda)$:

$$\frac{\sigma_i}{\sigma_i^2 + \lambda^2} = \frac{1}{\sigma_i} \cdot \frac{\sigma_i^2}{\sigma_i^2 + \lambda^2} = \frac{\phi_i(\lambda)}{\sigma_i} \Rightarrow y_i = \frac{\phi_i(\lambda)}{\sigma_i} u_i^T b \quad (10)$$

Writing $x_{\text{RLS}} = Vy = \sum_i y_i v_i$ as a sum then gives

$$x_{\text{RLS}} = Vy = \sum_{i=1}^n \frac{\sigma_i}{\sigma_i^2 + \lambda^2} (u_i^T b) v_i = \sum_{i=1}^n \phi_i(\lambda) \frac{u_i^T b}{\sigma_i} v_i \quad (11)$$

Compared with x_{LS} , Tikhonov adds the filter factor $\phi_i(\lambda)$, which keeps terms with large singular values and suppresses those with small ones. The parameter λ sets the transition.

- if $\sigma_i \gg \lambda$, then $\phi_i(\lambda) \approx 1$ and the term is kept almost unchanged;
- if $\sigma_i = \lambda$, then $\phi_i(\lambda) = \frac{1}{2}$;
- if $\sigma_i \ll \lambda$, then $\phi_i(\lambda) \approx 0$ and the term is damped away.

2.2 Truncated SVD as a filter

The filter-factor form is not specific to Tikhonov regularization. Any method that damps each term of x_{LS} can be written as

$$x = \sum_{i=1}^n \phi_i \frac{u_i^T b}{\sigma_i} v_i, \quad (12)$$

and methods differ only in their choice of ϕ_i . Where Tikhonov damps the terms smoothly, the truncated SVD simply cuts them off. We pick a threshold $\delta > 0$ and let k be the number of singular values larger than δ . The filter factors are then

$$\phi_i = \begin{cases} 1 & \text{if } i \leq k \\ 0 & \text{if } i > k \end{cases} \quad \begin{matrix} (\sigma_i > \delta) \\ (\sigma_i \leq \delta) \end{matrix}. \quad (13)$$

Substituting this in, only the first k terms survive:

$$x = \sum_{i=1}^k \frac{u_i^T b}{\sigma_i} v_i + \sum_{i=k+1}^n 0 \cdot \frac{u_i^T b}{\sigma_i} v_i = x_{\text{TLS}}. \quad (14)$$

The truncated least-squares solution is thus a filter method with a sharp cut-off at $\sigma_i = \delta$, instead of the smooth transition of Tikhonov regularization.

2.3 The generalized SVD

So far we have used $L = I$, which only penalizes the size of x . For a general L the SVD of A is no longer enough. Instead we use the generalized SVD (GSVD), which decomposes both matrices at once:

$$A = U \Sigma_A W^{-1}, \quad L = V \Sigma_L W^{-1}. \quad (15)$$

Here U and V are orthogonal, W is invertible, and Σ_A and Σ_L are diagonal. The ratios $\gamma_i = \alpha_i / \beta_i$ are the *generalized singular values* of (A, L) .

2.4 General-form regularization and the GSVD

We first characterize the solution of the general problem in Equation 2. Expanding its objective $f(x)$ as in the case $L = I$ but now with a general L gives

$$\begin{aligned} f(x) &= (Ax - b)^T (Ax - b) + \lambda^2 (Lx)^T (Lx) \\ &= x^T A^T A x - 2x^T A^T b + b^T b + \lambda^2 x^T L^T L x. \end{aligned} \quad (16)$$

Since f is convex, x is a minimizer exactly when $\nabla f(x) = 2(A^T A + \lambda^2 L^T L)x - 2A^T b = 0$. This is the normal equations for the general problem

$$(A^T A + \lambda^2 L^T L)x = A^T b. \quad (17)$$

The solution is unique when $A^T A + \lambda^2 L^T L$ is invertible. **behövs antagligen ett steg till här, när är den inverterbar?**

Instead of solving Equation 17 directly, we use the GSVD Equation 15, which shows what the regularization does to each component. Here

$$\Sigma_A = \text{diag}(\alpha_1, \alpha_2, \dots, \alpha_r, 0), \quad \Sigma_L = \text{diag}(\beta_1, \beta_2, \dots, \beta_r, 0), \quad (18)$$

and we assume all $\alpha_i > 0$. Substituting the GSVD into the expanded $f(x)$ gives

$$\begin{aligned} f(x) = & x^T W^{-T} \Sigma_A^T U^T U \Sigma_A W^{-1} x - 2x^T W^{-T} \Sigma_A^T U^T b + b^T b \\ & + \lambda^2 x^T W^{-T} \Sigma_L^T V^T V \Sigma_L W^{-1} x. \end{aligned} \quad (19)$$

With the change of variables $y = W^{-1}x$ this becomes

$$f = y^T (\Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L) y - 2y^T \Sigma_A^T U^T b + b^T b. \quad (20)$$

Since W is invertible, minimizing over x is the same as minimizing over y . Setting the gradient with respect to y to zero gives

$$\begin{aligned} \nabla f(y) = & 2(\Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L) y - 2\Sigma_A^T U^T b = 0 \\ \Rightarrow & \underbrace{(\Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L)}_B y = \Sigma_A^T U^T b. \end{aligned} \quad (21)$$

$$\begin{aligned} B := \Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L = & \begin{bmatrix} \alpha_1^2 & & & \\ & \alpha_2^2 & & \\ & & \ddots & \\ & & & \alpha_r^2 \\ & & & & 0 \end{bmatrix} + \lambda^2 \begin{bmatrix} \beta_1^2 & & & \\ & \beta_2^2 & & \\ & & \ddots & \\ & & & \beta_r^2 \\ & & & & 0 \end{bmatrix} \\ = & \begin{bmatrix} \alpha_1^2 + \lambda^2 \beta_1^2 & & & \\ & \alpha_2^2 + \lambda^2 \beta_2^2 & & \\ & & \ddots & \\ & & & \alpha_r^2 + \lambda^2 \beta_r^2 \\ & & & & 0 \end{bmatrix} \end{aligned} \quad (22)$$

Solving for y gives

$$\begin{aligned} y = & B^{-1} \Sigma_A^T U^T b \\ = & \begin{bmatrix} \frac{1}{\alpha_1^2 + \lambda^2 \beta_1^2} & & & \\ & \frac{1}{\alpha_2^2 + \lambda^2 \beta_2^2} & & \\ & & \ddots & \\ & & & \frac{1}{\alpha_n^2 + \lambda^2 \beta_n^2} \end{bmatrix} \begin{bmatrix} \alpha_1 & & & \\ & \alpha_2 & & \\ & & \ddots & \\ & & & \alpha_n \end{bmatrix} \begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_n^T \end{bmatrix} b \end{aligned} \quad (23)$$

Because B^{-1} and Σ_A are both diagonal, each component of y can be read off directly:

$$y_i = \frac{\alpha_i}{\alpha_i^2 + \lambda^2 \beta_i^2} u_i^T b, \quad i = 1, \dots, n, \quad (24)$$

and we recover the solution as $x = Wy$.

As for Tikhonov, we split the coefficient and introduce the filter factor $\phi_i(\lambda)$

$$\frac{\alpha_i}{\alpha_i^2 + \lambda^2 \beta_i^2} = \frac{1}{\alpha_i} \cdot \frac{\alpha_i^2}{\alpha_i^2 + \lambda^2 \beta_i^2} = \frac{\phi_i(\lambda)}{\alpha_i}, \quad \phi_i(\lambda) = \frac{\alpha_i^2}{\alpha_i^2 + \lambda^2 \beta_i^2} = \frac{\gamma_i^2}{\gamma_i^2 + \lambda^2}. \quad (25)$$

Writing $x = Wy = \sum_i y_i w_i$, where w_i are the columns of W , as a sum then gives

$$x = \sum_{i=1}^n \frac{\alpha_i}{\alpha_i^2 + \lambda^2 \beta_i^2} (u_i^T b) w_i = \sum_{i=1}^n \phi_i(\lambda) \frac{u_i^T b}{\alpha_i} w_i. \quad (26)$$

Here $\gamma_i = \alpha_i / \beta_i$ weighs how well A measures a component (α_i) against how much L penalizes it (β_i):

- if $\gamma_i \gg \lambda$, then $\phi_i(\lambda) \approx 1$ and the term is kept;
- if $\gamma_i \ll \lambda$, then $\phi_i(\lambda) \approx 0$ and the term is damped away.

A term is thus damped when α_i is small and β_i is large (since $\alpha_i^2 + \beta_i^2 = 1$, the two go together): A barely measures the component, so dividing by α_i would amplify noise, and L penalizes it strongly. With $L = I$ we damp the components that A measures weakly. With a general L we damp the components that A measures weakly *relative to how much L penalizes them*. Note that α_i and β_i are not the singular values of A and L separately; they come from the pair (A, L) in the common basis W . The choice of L thus decides which components count as unstable, for example removing oscillations while keeping smooth parts of the signal.

2.5 One-dimensional deblurring (Task 5)

3 Results

4 Discussion

References

A Appendix: Code